

- superimposition of direct current on the normal alternating current of a motor running at no load;
- application of flexible heaters to the test specimens.

5.4 Thermal ageing sub-cycle

The cold test objects (at room temperature) should be placed directly in preheated ovens, so as to subject them to a consistent thermal shock in each cycle. Likewise, the hot test objects should be removed from the ovens directly into room air, so as to subject them to uniform thermal shock on cooling as well as on heating.

It is recognized that some materials deteriorate more rapidly when the products of decomposition remain in contact with the insulation surface, whereas other materials deteriorate more rapidly when the decomposition products are continually removed. The same conditions of oven ventilation shall be maintained for both the candidate and the reference systems.

If in service the products of decomposition remain in contact with the insulation, as may be the situation in totally enclosed machines, the tests should then be designed so that the oven ventilation will not completely remove these decomposition products. Ideally, concentration of the decomposition products should not change with the ageing temperature but in practical testing this may not be realizable. The rate of replacement of air during thermal ageing shall be reported.

Depending on the test facilities available, the type of test objects employed and other factors, it may be desirable to use other methods of heating and of handling the products of decomposition.

In addition to thermal ageing, which is interrupted periodically for diagnostic testing so as to monitor the thermal degradation, thermo-mechanical deterioration of an insulation system may also be produced by the expansion and contraction of the assembly occurring during the temperature cycling.

6 Diagnostic sub-cycle

6.1 Conditioning sequence

Following each sub-cycle of thermal ageing, each specimen shall be subjected to mechanical and moisture conditioning procedures, followed by a voltage withstand and other diagnostic tests, as appropriate.

6.2 Mechanical conditioning

It is recommended that the mechanical stress applied be of the same general nature as would be experienced in service and of a severity comparable with the highest stresses or strains expected in normal service. The procedure for applying this stress may vary with each type of test object and kind of intended service.

A widely used method for applying mechanical stress is to mount each test object on a shake table and operate it for a period of 1 h with a 50 Hz or 60 Hz oscillating motion. Other methods, such as repeated impact and bending, are also used.

A start-stop or reversing duty cycle may also be used as a technique for mechanically stressing windings in actual machines. However, mechanical ageing may be introduced. Since this effect is more severe with increasing machine size, this factor shall be taken into account.

6.3 Moisture conditioning

Moisture is recognized in many cases to be a major cause of variation in the properties of electrical insulation. It may cause different types of insulation failure under electrical stress. The absorption of moisture by solid insulation has a gradual effect of increasing dielectric loss and reducing insulation resistance, and it may contribute to a change in electric strength. Moisture on insulation enhances the ability of a voltage test to detect cracks and porosity in the insulation.

Within the diagnostic sub-cycle it is common to apply a moisture test. In this test each test specimen is exposed to humidity with moisture deposition on the winding. Voltage should not be applied to the test specimens during this period.

A test of two days duration with visible moisture present on the insulation surfaces, being a more severe condition than is met in normal service, has won wide applicability. Experience has shown that an exposure time of at least 48 h is required for moisture to penetrate the winding so that the insulation resistance reaches a fairly stable level.

NOTE In case of fully sealed insulation systems, a water immersion test may be additionally needed for moisture sealed application.

6.4 Voltage tests

In order to check the condition of the specimens and determine when the end of test life has been reached, voltage is applied as a part of the selected diagnostic sub-cycle. The value and waveform of the voltage to be applied is stated in the subsequent subclauses of this standard, as e.g. given in 8.4.4, 9.4.3, 10.4.3, 11.4.3 and 12.4.3. When power-frequency voltage is specified, the frequency shall be in the range of approximately 49 Hz to 62 Hz.

The voltage may be applied from coil to frame, from coil to coil, from turn to turn or from wire to wire as appropriate. If a moisture test is used, the voltage test is applied at room temperature when the test specimens are still wet. Tap water quality shall thereby be used. The specimen surface may be wiped dry at the end connections.

In certain cases, the presence of surface moisture may prevent normal application of the voltage and in such cases the specimen surface may be wiped free of water droplets immediately before the voltage application.

Care shall be taken that unintended switching surges do not subject the insulation systems to transient surge voltages.

Any failure in any component of the insulation system constitutes failure of the entire test specimen and fixes the end of test life.

Failure in any of the voltage check tests is indicated by an unusual level of current. Localized heating or the presence of smoke may also indicate a failure. Minor spitting and surface sparking should be recorded but do not constitute a failure.

Test equipment shall be of sufficient capacity to produce and reveal a failure.

6.5 Other diagnostic tests

It may be desirable to take periodic, non-destructive measurements of insulation condition on some of the specimens during the course of the tests. Factors such as insulation resistance, loss tangent, and partial discharge are examples. By noting changes in these measurements and correlating them with time before failure occurs, it may be possible to learn something about the ageing process in the insulation.

Some other diagnostic tests may also be used to determine end of test life, either complementing the voltage tests or replacing them. An end-point criterion may be established for each diagnostic test, with suitable justification reported.

7 Reporting and functional evaluation of data from candidate and reference systems

7.1 General

The procedures to determine suitable end-point criteria and construct a graph of thermal endurance are given in 5.2 of IEC 60034-18-1.

For a full qualification test, the log mean life of each test object is plotted with its 90 % confidence limits against the inverse temperature, according to IEC 60216-1. The abscissa units are reciprocal absolute temperature (1/K) but are usually expressed in Celsius temperature. The ordinate units are expressed in hours. The result should be a semi-logarithmic plot showing straight lines for the candidate system and reference system and 90 % confidence intervals for their mean values at each test temperature.

IEC 60034-18-1, 5.2 recommends a general list of information to be recorded and included in the test report. Additional items may be reported where relevant. IEC 60216-5 provides a guide for complete statistical analysis of the results.

7.2 Determining qualification

7.2.1 Overview

The first step is to define the expected service life and thermal class of the candidate system, then to compare the performance of reference and candidate systems with respect to the qualification criteria given in Table 3. Caution is recommended when qualifying a candidate system for a different thermal class and/or service life, because of the assumptions implicit in the approach.

Before proceeding with the evaluation by comparison, it shall be established that the regression lines of the candidate and reference systems fit the data well (it is recommended that $R^2 \geq 0,98$) and there is no indication of any change of ageing mechanism within the range of test temperatures. If either of the regression lines is non-linear, refer to 7.2.6, where a simple test of linearity is described.

Table 3 – Conditions for qualification of candidate system

Case	Performance relative to reference system		Test temperatures (from Table 2)	Qualification criteria
	Class temperature (T_{class})	Expected service life		
A	Same	Same	Same	Confidence interval of candidate system overlaps or exceeds that of the reference system, at all test temperatures.
B	Same	Different	Same	Following the appropriate adjustments to the candidate system confidence limits (see description in the text for each case): 1. Confidence interval of candidate system shall overlap or exceed the confidence interval of the reference system. 2. The candidate system shows continually improving performance, i.e., the slope of its regression line is greater than or equal to the slope of reference system regression line.
C	Different	Same	Different	
D	Different	Different	Different	

7.2.2 Case A: Qualification for the same class temperature and same expected service life

To qualify the candidate system for the same class temperature and same expected service life (Table 3, Case A), the candidate system and reference system are tested using the same thermal ageing cycles. The candidate system is qualified if its confidence interval overlaps or exceeds that of the reference across the range of test temperatures. An example is shown in Figure 1, with candidate system “C” compared to reference system “R” and showing exceeding confidence intervals at all test temperatures. $T_{C,R}$ is the thermal class of the reference system.

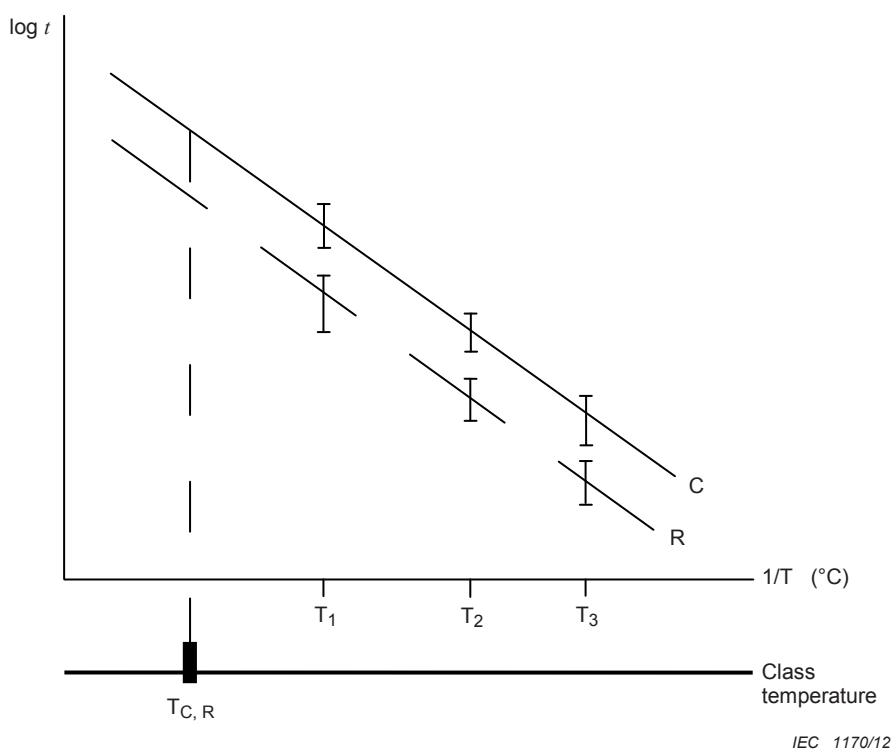


Figure 1 – Candidate system qualified for the same thermal class and the same expected service life

7.2.3 Case B: Qualification for the same class temperature and a different expected service life

To qualify the candidate system for the same class temperature and a different expected service life (Table 3, Case B), the candidate system and reference system are tested using the same thermal ageing cycles.

The candidate confidence limits at each temperature are shifted on the vertical axis by an amount equal to the agreed-upon change in service life, within the range $X_R/2$ to $2X_R$, where X_R is the life of the reference system at each temperature. The candidate system is qualified if its shifted confidence intervals overlap or exceed those of the reference system, and the candidate system shows a continually improving performance, i.e., the slope of its regression line is steeper than or equal to the slope of the reference system regression line.

Figure 2 shows an example of a candidate system assessed for qualification for the same thermal class, and an expected service life double that of the reference system. When the full

candidate system line is decreased by a factor of 2, its 90 % confidence limits overlap those of the reference. For simplicity, the shift on the vertical axis is shown for only the confidence limits at a single temperature, T_1 . Note that $T_{C,R}$ is the thermal class of the reference system.

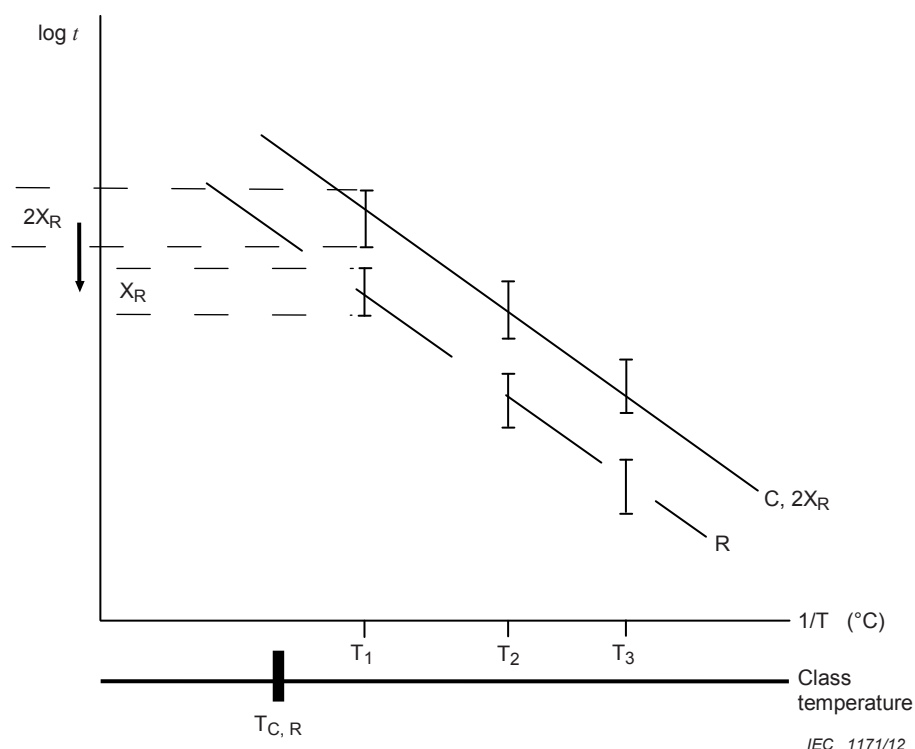


Figure 2 – Candidate system qualified for the same thermal class and different expected service life

If the candidate system regression line crosses that of the reference system within the range of measurement, it is qualified for a higher service life at $T_{C,R}$ only if it demonstrates improved performance compared to the reference system at $T_{C,R}$ by the required life interval.

If there exists the possibility of thermal overloading of the insulation system while in service, the qualification requirements for the candidate system should be evaluated as in case A, where the candidate system lifetime is equal to or better than that of the reference system across the test temperature range.

7.2.4 Case C: Qualification for a different class temperature and same expected service life

To qualify a candidate system for a different class temperature and the same expected service life (Table 3, Case C), the candidate system is tested using the ageing cycles that are appropriate to its intended thermal class. This approach is valid provided that the intended thermal class of the candidate is no more than one class higher or lower than the reference system thermal class. The lowest reference system test temperature should not be more than 25 K from its known class temperature, and the lowest test temperature of the candidate system should not be more than 25 K from its intended class temperature.

The candidate confidence limits at each temperature are shifted on the horizontal axis by an amount equal to the agreed-upon change in class temperature, where the intended thermal class of the candidate is no more than one higher or lower than the reference system thermal class. The candidate system is qualified if its shifted confidence interval overlaps or exceeds that of the reference system and the candidate system shows continually improving

performance, i.e., the slope of its regression line is steeper than or equal to the slope of reference system regression line.

Figure 3 shows an example of a candidate system assessed for qualification for the same expected service life at the next higher class temperature, where the confidence interval for the candidate tested using the cycles for a higher class temperature is shifted back to the class temperature of the reference. When the horizontal axis shift is completed, the confidence intervals overlap or exceed those of the reference system. $T_{C,R}$ is the thermal class of the reference system and T_{C+1} is the intended thermal class of the candidate.

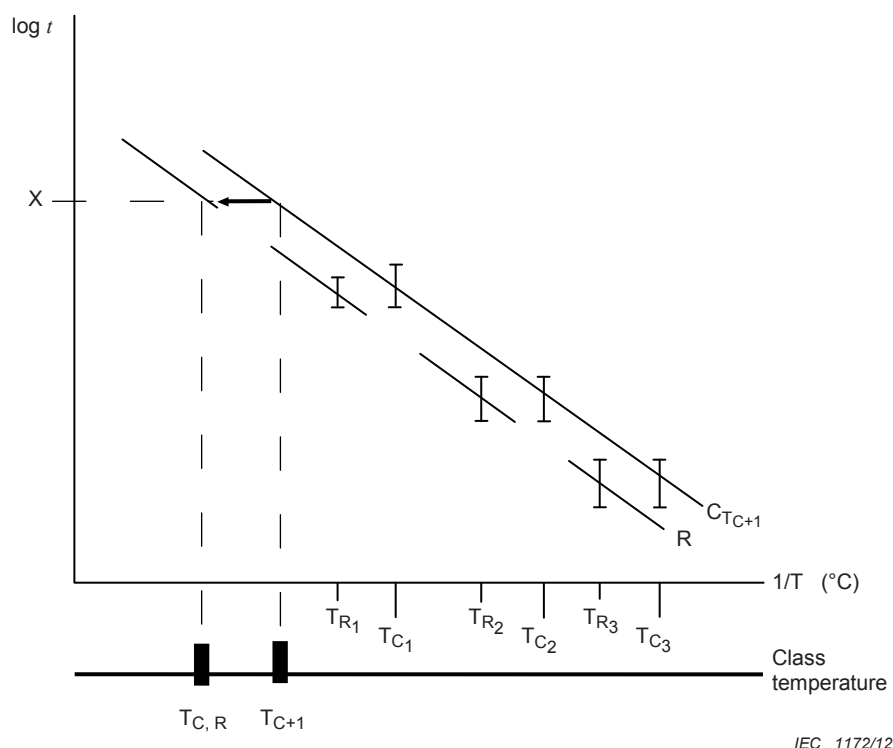


Figure 3 – Candidate system qualified for a different class temperature and the same expected service life

7.2.5 Case D: Qualification for a different class temperature and different expected service life

To qualify a candidate system for a different class temperature and a different expected service life (Table 3, Case D), the candidate system is tested using the ageing cycles that are appropriate to its intended thermal class. This approach is valid provided that the intended thermal class of the candidate is no more than one higher or lower than the reference system thermal class. The lowest reference system test temperature should not be more than 25 K from its known class temperature and the lowest test temperature of the candidate system should not be more than 25 K from its intended class temperature.

Qualification of the candidate system is determined by a shift of both the vertical and horizontal axes.

The candidate confidence limits at each temperature are shifted on the vertical axis by an amount equal to the agreed-upon change in service life, within the range $X_R/2$ to $2X_R$, where X_R is the life of the reference system at each temperature.

The candidate confidence limits at each temperature are then shifted on the horizontal axis by an amount equal to the agreed-upon change in class temperature, where the intended thermal

class of the candidate is no more than one higher or lower than the reference system thermal class. The candidate system is qualified if its shifted confidence interval overlaps or exceeds that of the reference system, and the candidate system shows continually improving performance, i.e., the slope of its regression line is steeper than or equal to the slope of reference system regression line.

Figure 4 shows an example of a candidate system assessed for qualification for double the expected service life of the reference, at the next higher class temperature than the reference. For simplicity, the shift on the vertical axis is shown for only the confidence limits at a single temperature, T_1 . The same candidate system is also assessed for qualification at a higher class temperature, where the confidence interval for the candidate tested using the cycles for a higher class temperature is shifted back to the class temperature of the reference, where $T_{C,R}$ is the thermal class of the reference system and T_{C+1} is the intended thermal class of the candidate.

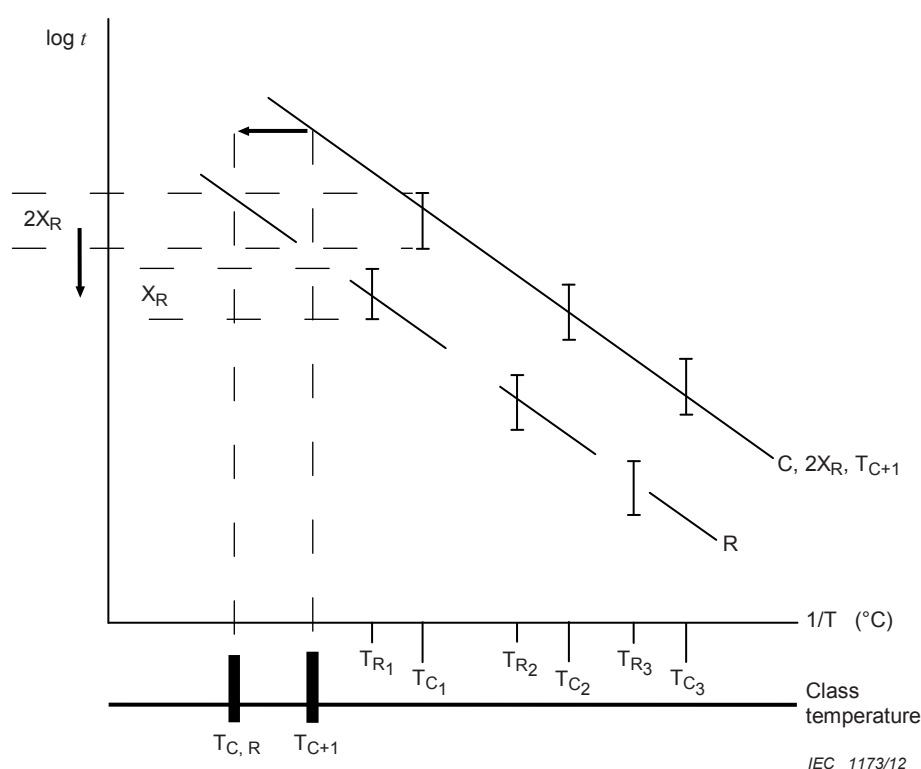


Figure 4 – Candidate system qualified for a different service life and different thermal class from the reference

7.2.6 Non-linearity of regression lines

The candidate and reference systems may respond differently to the combination of ageing factors, resulting in curved regression lines. A slight bend in the graph indicates that more than one chemical process or failure mechanism influences thermal ageing. If a straight line cannot be drawn within the tolerance bars of all the points, the data suggest that there is a significant change in the principal ageing mechanism within the range of test temperatures. Confirmation of the curve by obtaining an additional test point at a lower or intermediate temperature is recommended.

7.2.7 Reduced evaluation

For a reduced evaluation, one test object is tested at a single time-temperature cycle within the temperature range used to produce the reference line. In this case, the log mean life of

the test object is similarly plotted, along with its 90 % confidence limits, against the reference system line.

This approach is not as rigorous or complete as that of a full qualification, and therefore it is reserved for evaluation of minor changes to the insulation system, i.e., those not expected to affect significantly the system's endurance under thermal stress conditions.

The candidate system is qualified by reduced evaluation if the 90 % confidence limits of the point for the test object overlaps or exceeds the confidence interval of the reference system.

8 Procedure 1: Motorette test procedure

8.1 General

8.1.1 Test object definition

Test objects may be actual machines, machine components or models. This procedure, using motorettes as test objects, shall be referred to as IEC 60034-18-21, Procedure 1.

8.1.2 Test procedure

This thermal endurance test procedure consists of several cycles. Each cycle consists of:

- a thermal ageing sub-cycle;
- a diagnostic sub-cycle which includes a mechanical test, a moisture test with test-specimen cooling and a voltage test, performed in that order.

8.2 Test objects

8.2.1 Construction of test objects

The test object in this procedure, designated motorette, models the insulation system to be tested.

The motorette shall be made to embody all of the essential elements and should be as nearly as possible representative of a complete winding insulation system.

An example of a motorette used to test wire-wound winding insulation is described in Annex A. The motorette simulates a wire-wound distributed or concentrated winding of a slotted stator structure.

For the case of qualifying concentrated windings, the pole winding motorettes of Annex B should preferably be used.

8.2.2 Number of test objects

At least ten motorettes should be tested at each ageing temperature, for each insulation system.

8.2.3 Quality assurance tests

Before the first thermal ageing sub-cycle is started, the following quality assurance tests shall be performed:

- visual inspection of the test objects;
- voltage tests according to IEC 60034-1;
- 400 V a.c. conductor-to-conductor test with 50 mA circuit breaker to detect failure.

8.2.4 Initial diagnostic tests

Each completed test object shall be subjected to the diagnostic tests of Clause 6, before starting the first thermal ageing sub-cycle.

8.3 Thermal ageing sub-cycle

8.3.1 Ageing temperatures and sub-cycle lengths

The procedures given in 5.2 shall be followed.

8.3.2 Means of heating

Ageing ovens according to 5.3 shall be used.

8.3.3 Ageing procedure

The motorettes shall be placed directly into the hot ageing oven at the beginning of the ageing sub-cycle, and removed from the oven directly to room temperature air at the end of the sub-cycle.

In order to diminish the effects of differences in actual ageing temperatures between individual motorettes, the locations of the motorettes in the ageing oven should be randomized in successive heat ageing sub-cycles.

8.4 Diagnostic sub-cycle

8.4.1 General

For performing the diagnostic sub-cycles, the procedures and explanations given in Clause 6 are applied.

8.4.2 Mechanical conditioning

Following each sub-cycle of thermal ageing and after cooling to room temperature, each motorette is subjected to mechanical stress on a shake table for a period of 1 h.

The preferred amplitude of the vibration corresponds to an acceleration of 1,5 g (0,2 mm peak-to-peak amplitude at 60 Hz or 0,3 mm at 50 Hz). If the principle of service-related stresses (see 6.2) leads to a larger vibration amplitude, it shall be used and reported.

The motorettes are thereby mounted so that the motion occurs at right angles to the plane of the coils so that the coil ends are excited to vibrate as they would under radial end-winding forces in an actual motor. This vibration test is made at room temperature and without any applied voltage.

8.4.3 Moisture conditioning

Moisture conditioning shall be performed for at least 48 h, according to 6.3. No voltage is applied during the moisture exposure.

The principle of cooled test objects shall be used. See Clause C.2 of Annex C. The motorettes shall be at a 15 °C to 35 °C temperature range. The actual motorette temperatures shall be reported. No voltage is applied to the test objects during this test. See Annex C for examples of equipment for such tests. Alternatively, climatic chambers can also be utilized.

8.4.4 Voltage test

In order to check the condition of the test specimens and determine when the end of test life has been reached, power-frequency voltage is applied after each successive exposure to

moisture, as given in Table 4. The diagnostic voltage test is carried out throughout the thermal ageing sub-cycle.

The test voltage to be used to frame and between coils should correspond to the upper limit of the voltage range for which that insulation system is intended. A voltage other than 660 V may be employed in order, for example, to permit the use of much test data taken at 600 V. Other test voltages may be used for end-point determination based on test experience as long as these voltages are maintained consistently for both the reference and the candidate systems. Deviations from values given in Table 4 shall be reported.

Table 4 – Test voltages

Rated voltage in service	Recommended power-frequency voltage for testing V (r.m.s. value)		
	To frame	Between coils	Between conductors ^a
< 400	400	400	110
401 – 690	690	690	110
> 690	$2 U_N$	$2 U_N$	110
^a Range of acceptable voltages; however, the value chosen should be used consistently.			

The voltages are applied for a period of 10 min after the test specimens are kept 48 h in the equipment at 95 % RH to 98 % RH, wet with moisture, but without any droplets. The applied voltage is held successively, each time for 10 min using appropriate circuitry, first between the parallel wound conductors, then from coil to coil, and finally from all coils to frame. It is suggested that surge protectors be included in the test circuit to eliminate unintended high-voltage spikes.

Experience indicates that the characteristics of the circuit breakers used to detect failure are very important. See Clause A.3 of Annex A.

8.4.5 Other diagnostic tests

Other diagnostic tests may be performed according to 6.5.

8.5 Analyzing, reporting and classification

The procedures given in Clause 7 shall be followed.

9 Procedure 2: Motor test procedure

9.1 General

9.1.1 Test object definition

Test objects may be actual machines, machine components or models. This procedure, using actual motors as test objects, shall be referred to as IEC 60034-18-21, Procedure 2.

9.1.2 Test procedure

This thermal endurance test procedure consists of several cycles. Each cycle consists of:

- a thermal ageing sub-cycle;
- a diagnostic test sub-cycle, which may include a moisture test. Voltage is applied continuously during the running of the motor and is also a diagnostic factor.

Greater thermomechanical stress and higher concentration of the products of decomposition occur during tests at higher than actual temperature. Also, it is recognized that failures from abnormally high mechanical stress or voltages are generally of a different character from those failures which are produced in long service.

Due to variations in control of key test parameters, manufacturing processes and methods of testing motors, it is exceedingly difficult to compare motor tests of one facility to those of another. It is the intent of this procedure to compare motor insulation systems within one manufacturing and one testing facility.

Even though actual motors are tested, the results may not be used to determine endurance time in actual service in an absolute sense. The tests may be used as a means of classification only by comparing insulation systems.

9.2 Test objects

9.2.1 Construction of test objects

The test objects are complete motors. A motor may be modified for the test to increase its mechanical life. To increase its temperature rise various techniques may be employed provided no changes are made in the insulation system and its immediate environment.

In the tests on actual motors the dimensions of components and the manufacturing processes of winding and shaping do affect the test results. Therefore, the manufacturing processes should be those used or contemplated for use in normal production.

9.2.2 Number of test objects

At least five motors should be tested at each ageing temperature for each insulation system.

9.2.3 Quality assurance tests

Before the first thermal ageing sub-cycle is started, the following quality assurance tests shall be performed:

- visual inspection before assembly of the motors;
- voltage tests according to IEC 60034-1.

9.2.4 Initial diagnostic tests

Each completed test object shall be subjected to the diagnostic tests of 9.4, before starting the first thermal ageing sub-cycle.

9.3 Thermal ageing sub-cycle

9.3.1 Ageing temperatures and sub-cycle lengths

The procedures given in 5.2 shall be followed.

Test temperatures shall be measured by the resistance method. Thermocouples may be installed for purposes of control. The temperature should be controlled to the accuracy prescribed in 5.3 after the thermal ageing temperature is reached. If the temperature of any one motor deviates appreciably from the average for the group being run at a common temperature it should be so reported and taken into account in the analysis of the data.

9.3.2 Means of heating

The mode of heat generation is dictated by the type of motor being used in the test and the laboratory equipment available. Higher than normal winding temperatures can be obtained by increasing motor losses by such means as enlarging the air gap, starting and reversing each

motor, superimposition of direct current on the normal alternating current, or by increasing the temperature of the air surrounding the motor. For temperature regulation during the heat ageing portion of the cycle, the motors may be run at normal voltage and frequency with an electrical control which automatically starts and stops or reverses the direction of rotation of the motors at intervals. Other acceptable means of temperature control include automatic voltage variation, adjustment of the surrounding air temperature, or combinations thereof.

The means of heating shall be described in detail in the test report.

Single-phase motors shall have at least 250 start-stop operations each day of the heat ageing portion of the cycle. The starter winding of a single-phase motor normally operates at a much higher current density than the main winding during starting. The number of starts should be chosen to ensure a temperature of 10 K to 30 K higher than the main winding.

Polyphase motors shall have at least 1 000 starts or reversals each day of the heat ageing portion of the cycle. Often the electrical loss during reversal is used to maintain the elevated temperatures, in which case the number of reversals may greatly exceed 1 000 per day. At the highest temperature test the total time of exposure is relatively short which results in a relatively low number of reversals during the life of the test. At the lowest temperature, the time of exposure can be 16 to 20 times as long as that of the highest level. A wide variation in total number of starts would affect the slope of the time-temperature curve within a cycle. Thus, it is recommended that the number of reversals at the low temperature be no greater than twice those at the high temperature. Ideally, an equal number of reversals at each temperature should be sought.

9.3.3 Ageing procedure

Motors are run during the thermal ageing cycle as described in 9.3.2. The heating-up time is to be considered as part of the thermal ageing period while the cooling-down time is not. At the end of the ageing sub-cycle, motors are allowed to cool to room temperature before starting the diagnostic sub-cycle. The cooling rate may be increased by running the motors at no-load for a time, with unrestricted ventilation if the machines are open-ventilated.

9.3.4 Mechanical stresses during the thermal ageing sub-cycle

Mechanical stress is obtained in tests on actual motors by the normal vibration of the motor running and with starts or reversals, or both. There is mechanical shock from starting or reversing. The vibration amplitude at twice the line frequency can be increased by enlarging the air gap. Vibration is a function of force and the larger air gap generally decreases the vibration rather than increasing the vibration. Larger forces are present in the windings as a result of the high currents during starting and reversing of the motors. In a test, these mechanical forces occur at elevated temperatures.

The driving method and vibration level should be reported.

The test motors should either be solidly mounted or mounted on shock pads that will give a uniform amount of shock to all motors. The mounting method shall be reported. The test objects containing the candidate insulation system shall be mounted in the same way as the test objects containing the reference system.

9.4 Diagnostic sub-cycle

9.4.1 Mechanical conditioning

Following each sub-cycle of thermal ageing, after cooling to room temperature, each motor is subjected to mechanical stress, for example on a shake table, for a period of 1 h.

The preferred amplitude of the vibration corresponds to an acceleration of 1,5 g (0,2 mm peak-to-peak amplitude at 60 Hz or 0,3 mm at 50 Hz). If the principle of service-related stresses (see 6.2) leads to a larger vibration amplitude, it shall be used and reported.

9.4.2 Moisture conditioning

Moisture conditioning shall be performed for at least 48 h, according to 6.3. No voltage is applied during the moisture exposure.

For totally enclosed machines (degrees of protection IP44 or more) and for d.c. machines a moisture test is not mandatory because it can be impracticable.

Moisture shall be visible on the windings as droplets, without puddles, during the moisture test. To ensure visible condensation, the insulation system should be at a lower temperature than the dew point of the surrounding moisture-laden atmosphere at all times. The preferred method of meeting this requirement is by the use of a condensation test chamber with cooled test objects described in Clause C.2 of Annex C.

However, larger motors can be difficult to move and difficult to support in equipment for a moisture test or such equipment may not be available. Other methods of applying moisture include: placing an enclosing hood around the motor, or using a conventional humidity cabinet or a fog chamber.

If totally enclosed machines are to be tested, end bells or the covers of terminal boxes should be removed, or openings should be provided in the enclosures for the moisture exposure.

9.4.3 Voltage withstand test

In order to check the condition of the test specimens and determine when the end of test life has been reached, power-frequency voltage is applied after each successive exposure to moisture. Typical test voltages to be used are shown in Table 4 with their related test guidelines in 6.4. The diagnostic voltage test is carried out throughout the thermal ageing sub-cycle.

The motors should be started and run immediately after the moisture test while the windings are still wet. For machines that have to be reassembled prior to running, a power-frequency high potential test should be applied at the highest rated voltage from windings to frame for 10 min while wet before assembly. During at least part of the thermal ageing subcycle the motors are to be run at their highest rated nameplate voltage. A power source earthed through a current limiting impedance should be used and the motor frame should be earthed so that voltage stresses are present during the entire thermal ageing portion of the cycle. A detection circuit for current to frame should be used to detect when insulation to the frame has failed. The end point of the motor life in these tests is fixed by the electrical failure of its winding insulation, under a rated applied voltage. Indiscriminate starting in either direction of the rotation of a single-phase motor can indicate failure of the starting winding.

It is suggested that surge protectors be included in the test circuit to eliminate unintended high-voltage spikes.

9.4.4 Other diagnostic tests

Other diagnostic tests may be performed according to 6.5.

The motors may be given a repeated surge comparison test applied to each winding or phase of the motor in sequence. Since surge tests also stress frame insulation, no voltage shall be used which is higher than the crest of the frame test voltage specified in IEC 60034-1.